Bounded recursion: every term rule has a unique ordinal-length approximation built by iterating it

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: uniqueness by agreement-on-overlap, the empty/successor/limit constructions, and the final induction closure sentence are all verbatim in the span with the successor-induction principle cited by name from the ordinals unit.

Editorial note. The label transrecursionfun names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Simple transfinite induction. Unit olp-set-theory-ordinals, \ollabel{simpletransrecursion} (zero/successor/limit induction principle, source line 934 of that unit). Existence part (proof lines 719-735): the induction on ordinals delta <= alpha with the three closure cases.
- Unions of coherent functions. Set-theoretic background of the unit: union of functions agreeing on overlaps is a function. Limit step at lines 731-735: g = union of the delta-approximations, well-defined since the approximations agree.

Source reference. Original reference simpletransrecursion: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 948–948. Named cross-unit citation: the target label is staged in the corpus under unit olp-set-theory-ordinals at source line 948; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Bounded Recursion)**.** *For any term $\tau(x)$ and any ordinal α, there is a unique α-approximation for τ.*

Proof. We will show that, for any $\gamma \leq \alpha$, there is a unique γ-approximation.

We first establish uniqueness. Let g and h (respectively) be γ- and δ-approximations. A transfinite induction on their arguments shows that $g(\beta) = h(\beta)$ for any $\beta \in \operatorname{dom} g \cap \operatorname{dom} h = \gamma \cap \delta = \min(\gamma, \delta)$. So our approximations are unique (if they exist), and agree on all values.

To establish existence, we now use a simple transfinite induction (simple-transrecursion) on ordinals $\delta \leq \alpha$.

The empty function is trivially an $\emptyset$-approximation.

If g is a γ-approximation, then $g \cup \{\langle \gamma, \tau(g) \rangle\}$ is a $+1\gamma$-approximation.

If γ is a limit ordinal and g_δ is a δ-approximation for all $\delta < \gamma$, let $g = \bigcup_{\delta \in \gamma} g_\delta$. This is a function, since our various g_δs agree on all values. And if $\delta \in \gamma$ then $g(\delta) = g_{+1\delta}(\delta) = \tau(g_{+1\delta} \restriction \delta) = \tau(g \restriction \delta)$.

This completes the proof by transfinite induction. $\square$